

Intermediate Algebra Test IV

Name
Date
Course

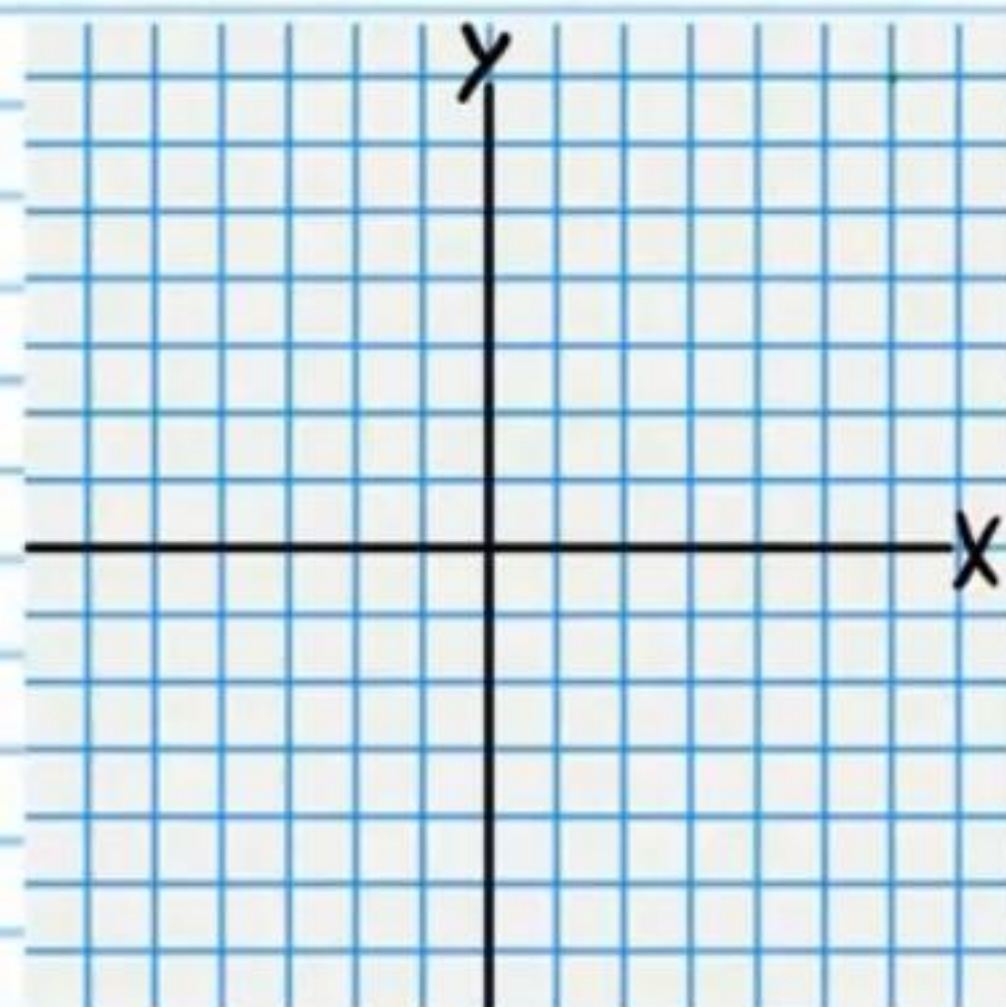
- I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- II) Show all work.
- III) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for this test.
- V) Each problem is worth four points for a total of 100 points.

Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation.

① $y+3 = |x+4|$

Parent Relation:

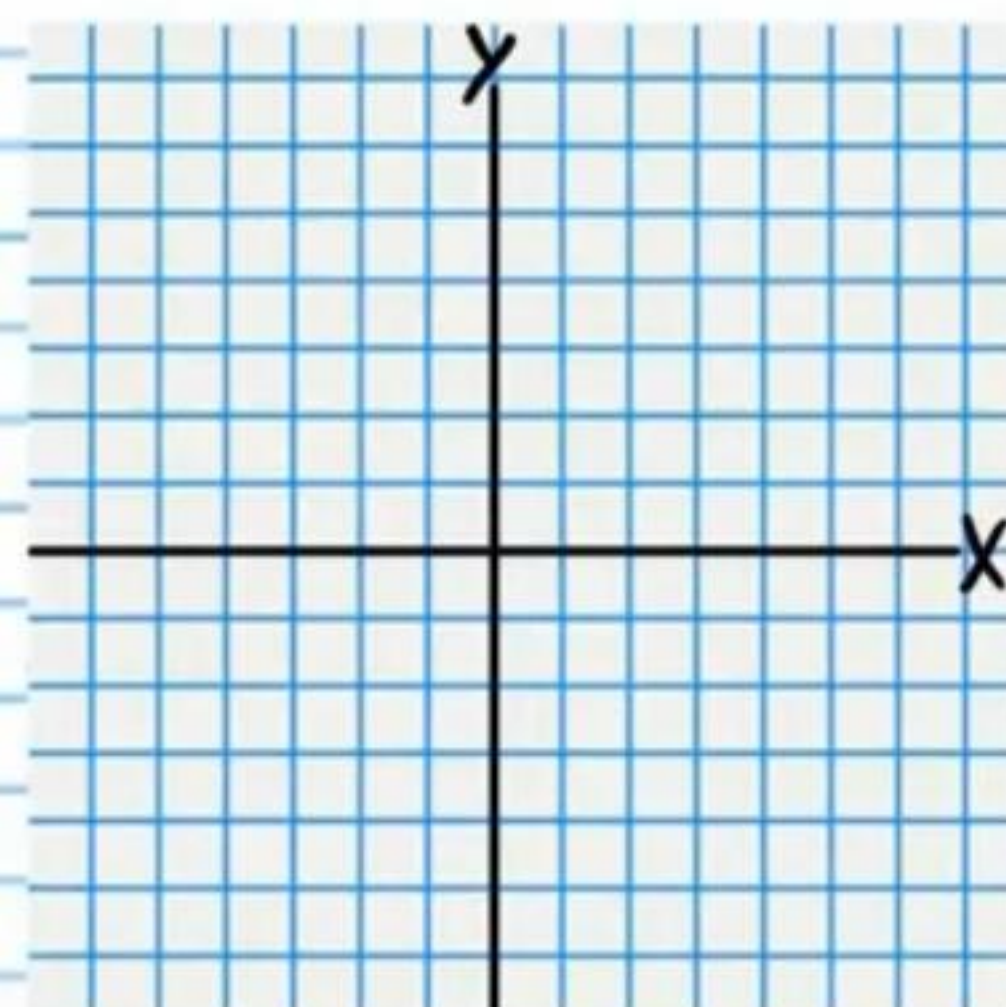
Parent Equation:



② $x^2 = 2y$

Parent Relation:

Parent Equation:



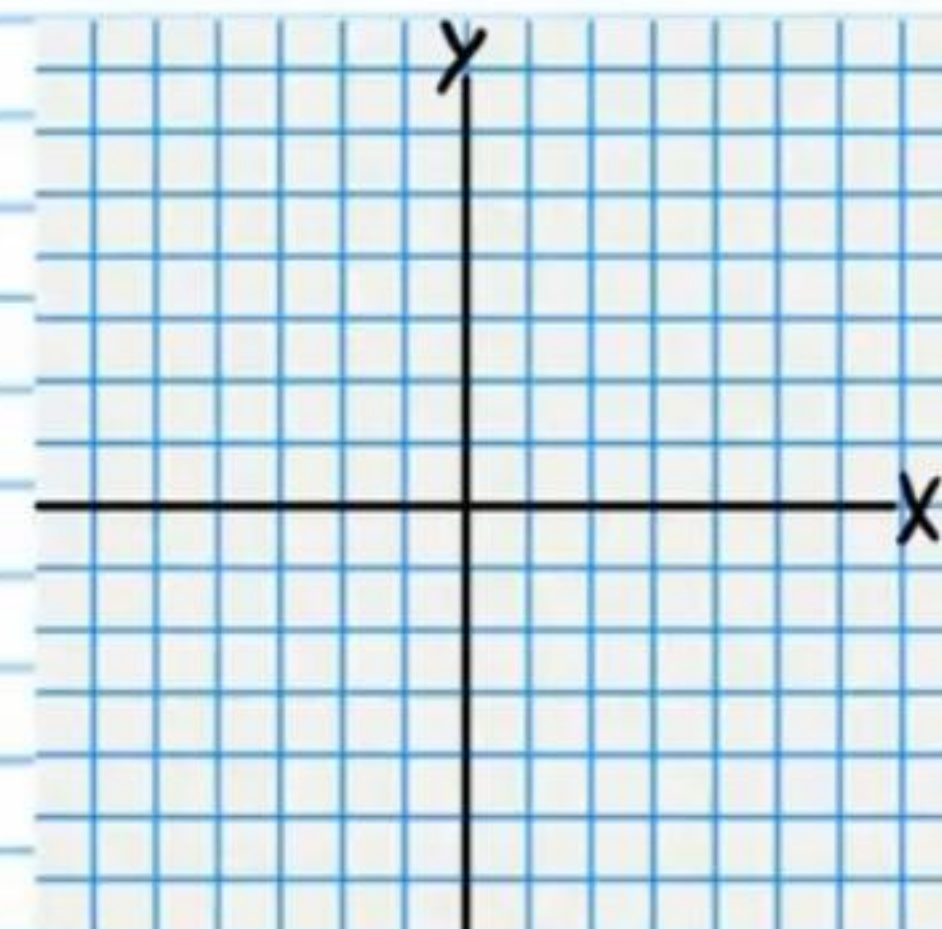
Write the parent relation and the parent equation of each relation below and use this information to graph the modified equation. Draw the asymptote(s) on the graph as dotted lines and write their equation(s). If applicable, identify the type of dilation involved.

③ $-y = 2^x$

Parent Relation:

Parent Equation:

Asymptote Equation:



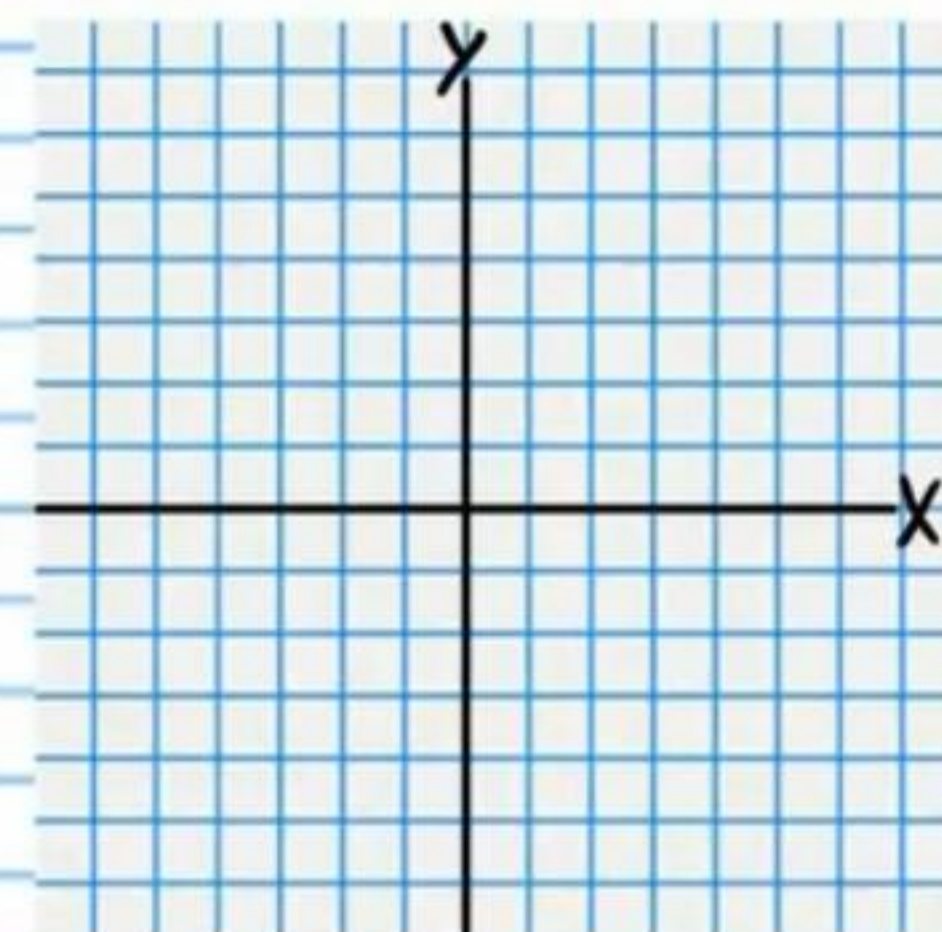
④ $\log_4\left(\frac{x}{3}\right) = y$

Parent Relation:

Parent Equation:

Dilation:

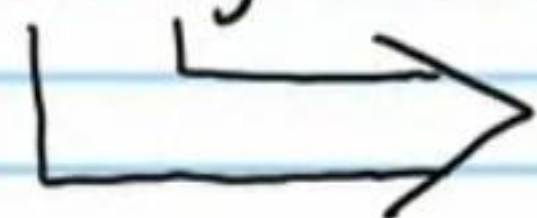
Asymptote Equations:



Change the form of each equation below so that it can be graphed with the transformation method. Draw the parent graph. Then draw each transformation, and write the equation for each graph. Draw any asymptotes as dotted lines.

⑤ $y = \frac{-1}{x}$

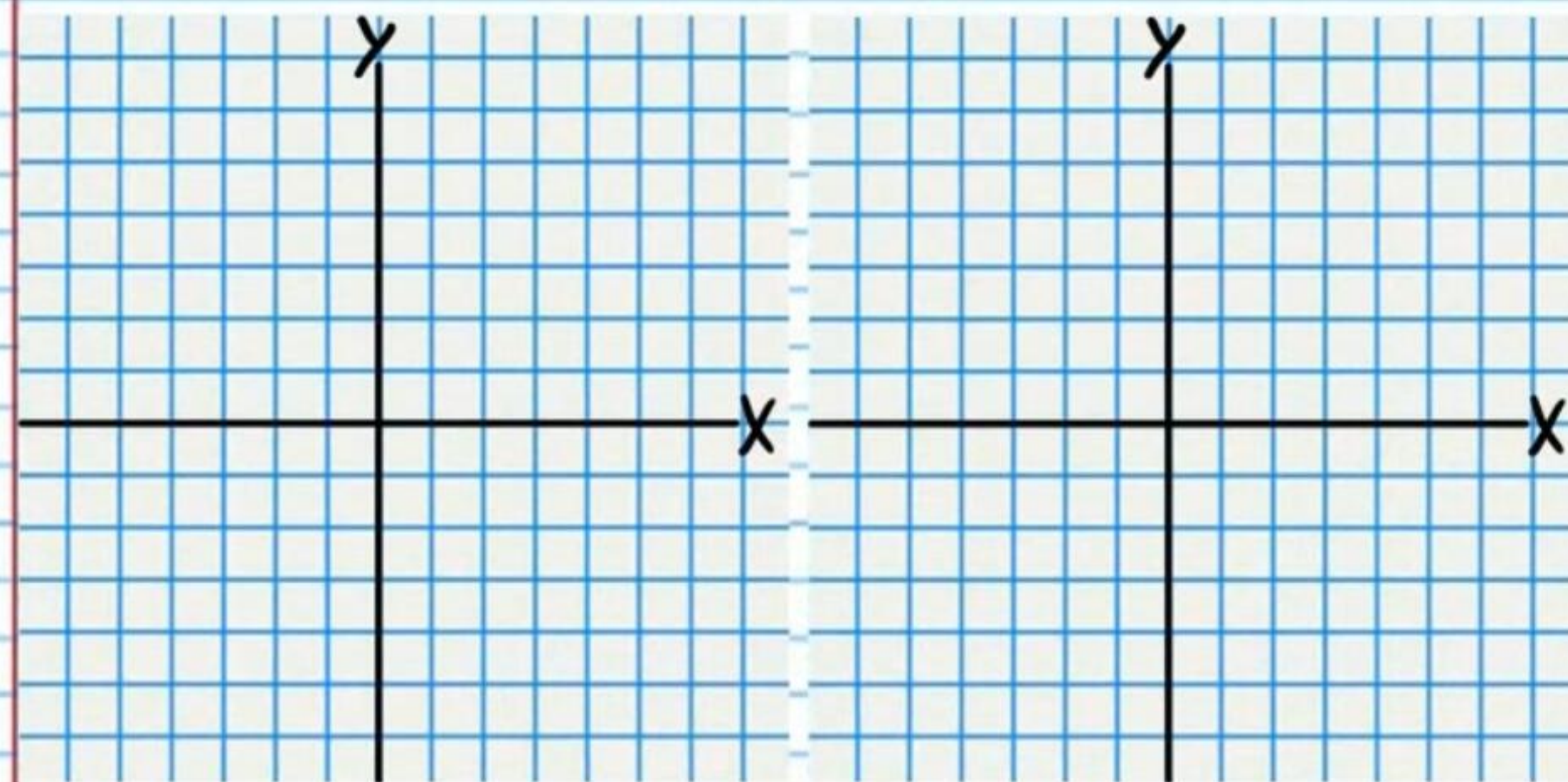
Change the form of the equation.



Altered Form:

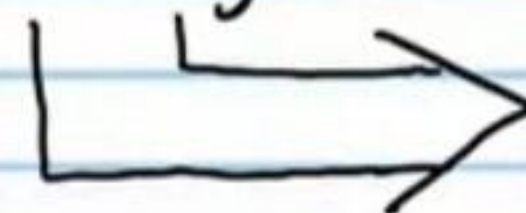
Parent Relation

Transformation



⑥ $y = -4x^3$

Change the form of the equation.

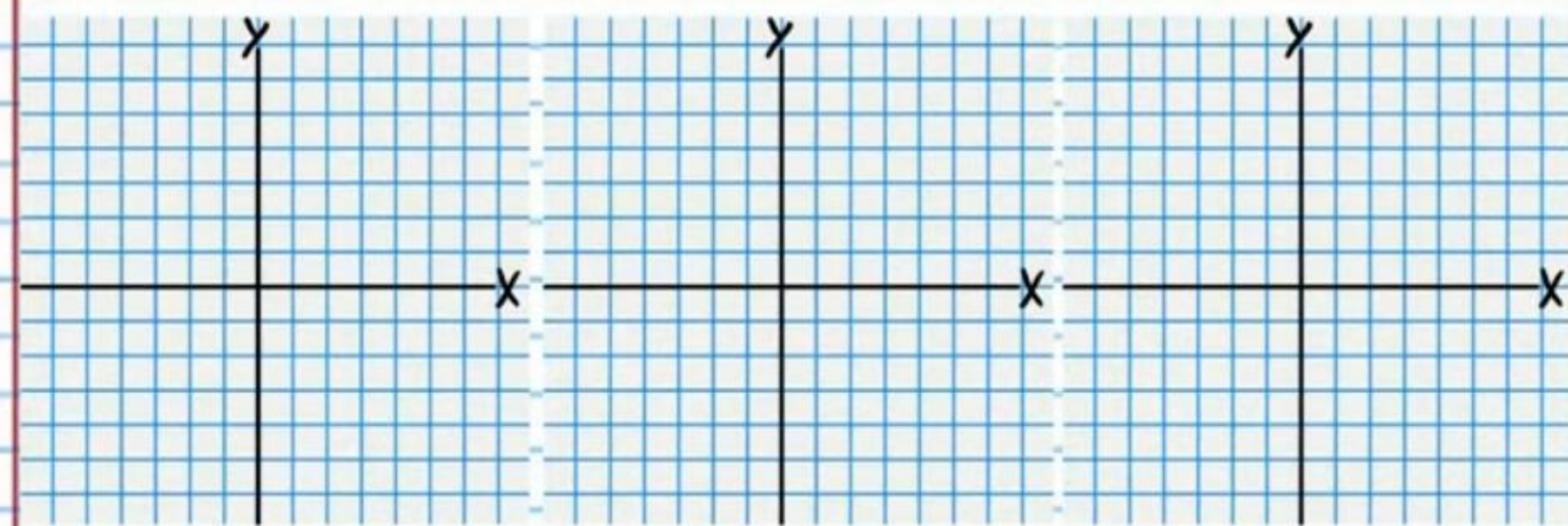


Altered Form:

Parent Relation

Transformation #1

Transformation #2



⑦ In the tables below, x represents the inputs and y represents the outputs. Circle the tables that could not represent functions.

x	y
-2	4
-1	5
0	10
1	12
2	9
2	-12

x	y
8	8
5	5
-2	14
0	5
2	2
8	7

x	y
3	15
2	10
1	8
0	10
-1	12
-2	14

x	y
7	0
10	0
9	0
4	0
2	4
1	87

⑧ If $f(x) = \sqrt{x^3+1}$, fill in the table below.

x	$f(x)$
-1	
1	
2	
3	

⑨ Given the function below, find the values.

$$f(x) = \begin{cases} 3^{x+2} + 2, & -10 \leq x \leq -3 \\ x^2 + x, & -3 < x \leq 0 \end{cases}$$

i) $f(-2)$

iii) $f(1)$

ii) $f(0)$

iv) $f(-3)$

⑩ If $f(x) = \log_2 x$ and $g(x) = \sqrt{x}$, find the following:

i) $(f-g)(x)$

ii) $(f \circ g)(4)$

⑪ Write the domain of the function below in interval notation.

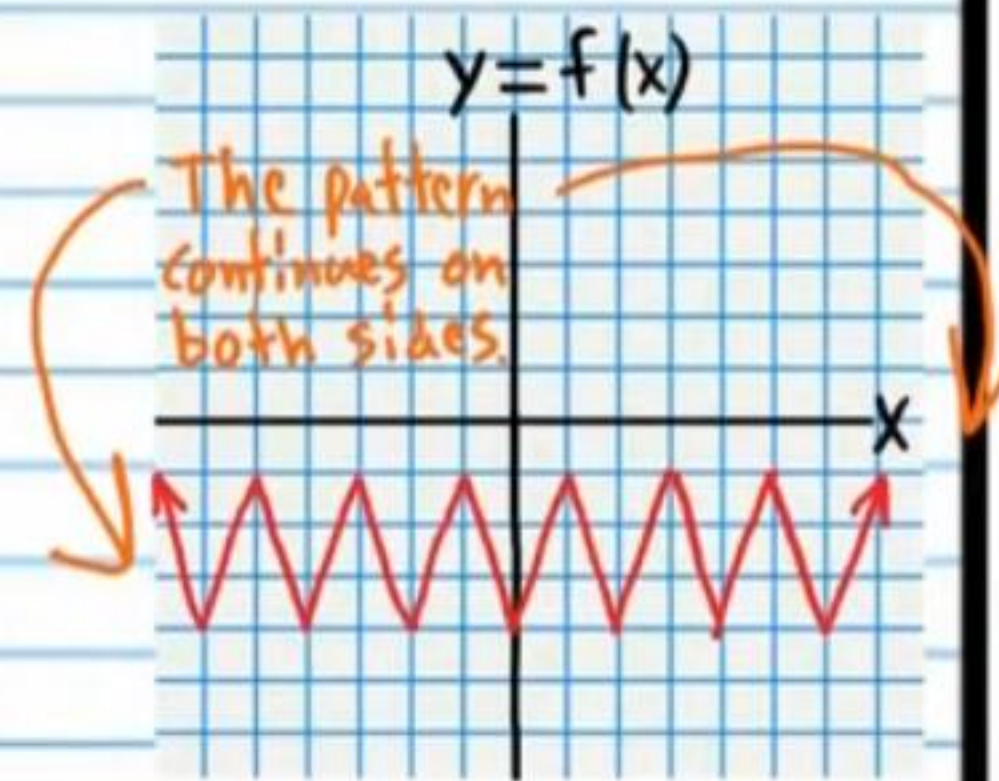
$$h(x) = \begin{cases} x-18, & -5 < x < 2 \\ 4x+6, & 8 \leq x \leq 100 \end{cases}$$

⑫ Write the domain of the function in problem #11 in set notation.

- ⑬ The graph of a function is written below. Write the domain and range of the function in set notation.

Domain

Range



- ⑭ If y is inversely proportional to x , and $y=10$ when $x=10$, find y when $x=50$.

- ⑮ If a 0.7% interest rate is compounded continuously for 11 years, and the initial investment is \$623, how much would the investment be worth?

- ⑯ If Claire deposits \$5000, paying an annual interest rate of 3.7% compounded daily, and his investment grows to \$8400, how much time passed? Assume that he made no withdrawals or additional deposits.

Solve. Round to the nearest thousandth.

(17) $3 = e^{2x} \cdot e^{4x}$

(18) $-1 = 2 - \frac{\log x}{2}$

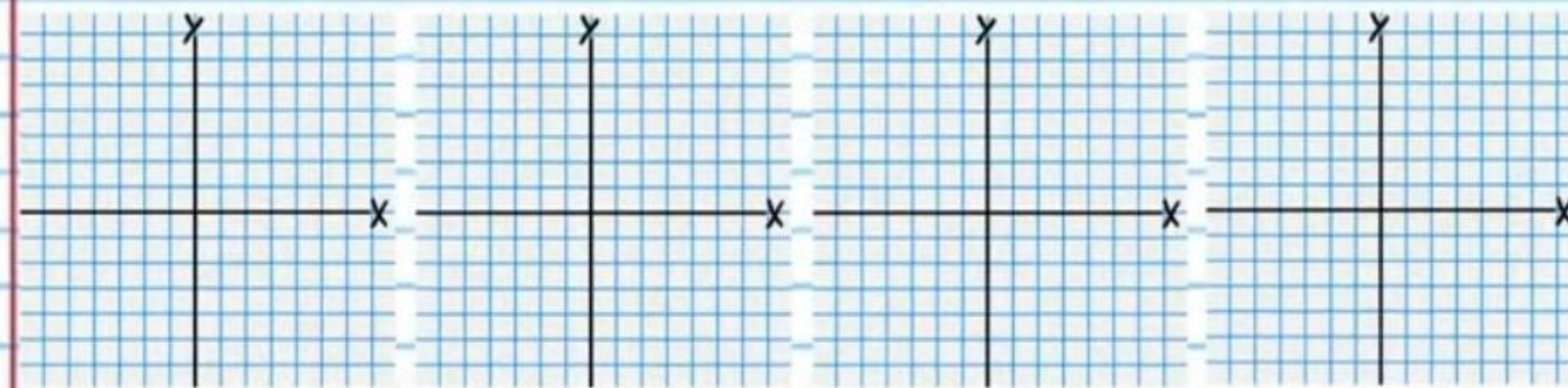
(19) What is the length of a disturbance on a seismometer 100 kilometers from the epicenter of an earthquake with magnitude 5.6?

(20) Write the domain of the function below in set notation. You may use the grids below if necessary.

(21) Write the range of the function below in interval notation. You may use the grids below if necessary.

$f(x) = \frac{\log_2(x-1)}{-4} + 3$

Parent Function Transformation #1 Transformation #2 Final Transformation

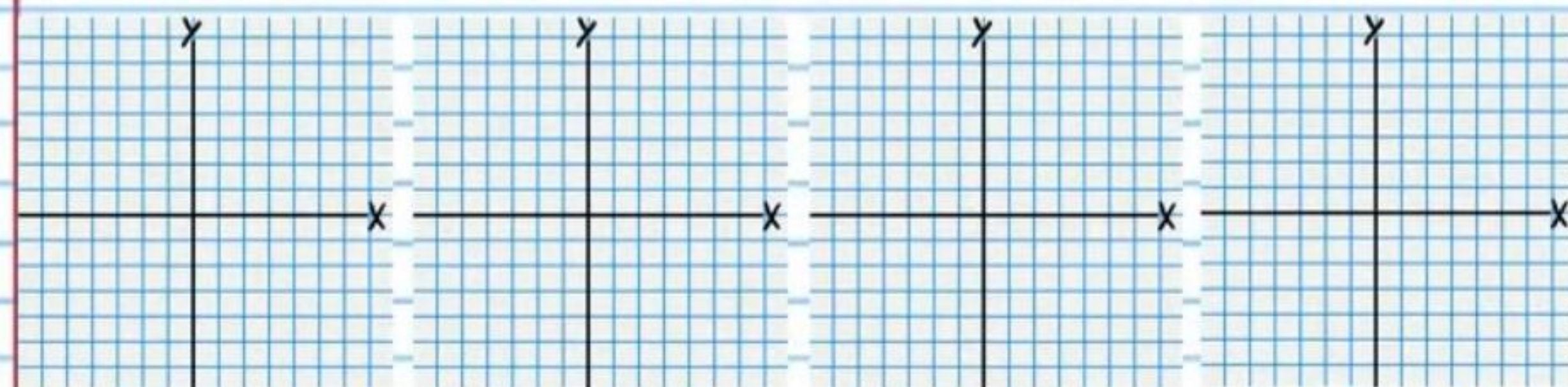


The transformation properties can be used with each pathway at the right to graph the corresponding equation. But in most of these pathways, expressions must be converted into a different form in order to use the transformation properties in the way that they are written. Choose the pathway(s) in which no modification of expressions is required. Then, use your chosen pathway to graph the corresponding equation.

②② $\sqrt{-x+3} = 2y+1$

Pathway #1	Pathway #2	Pathway #3
$\sqrt{x} = y$	$\sqrt{x} = y$	$\sqrt{x} = y$
$\sqrt{x+3} = y+1$	$\sqrt{x+3} = y+1$	$\sqrt{x} = 2y$
$\sqrt{x+3} = 2y+1$	$\sqrt{-x+3} = y+1$	$\sqrt{-x} = 2y$
$\sqrt{-x+3} = 2y+1$	$\sqrt{-x+3} = 2y+1$	$\sqrt{-x+3} = 2y+1$
Pathway #4	Pathway #5	Pathway #6
$\sqrt{x} = y$	$\sqrt{x} = y$	$\sqrt{x} = y$
$\sqrt{x} = 2y$	$\sqrt{-x} = y$	$\sqrt{-x} = y$
$\sqrt{x+3} = 2y+1$	$\sqrt{-x} = 2y$	$\sqrt{-x+3} = y+1$
$\sqrt{-x+3} = 2y+1$	$\sqrt{-x+3} = 2y+1$	$\sqrt{-x+3} = 2y+1$

Parent Relation Transformation #1 Transformation #2 Final Transformation

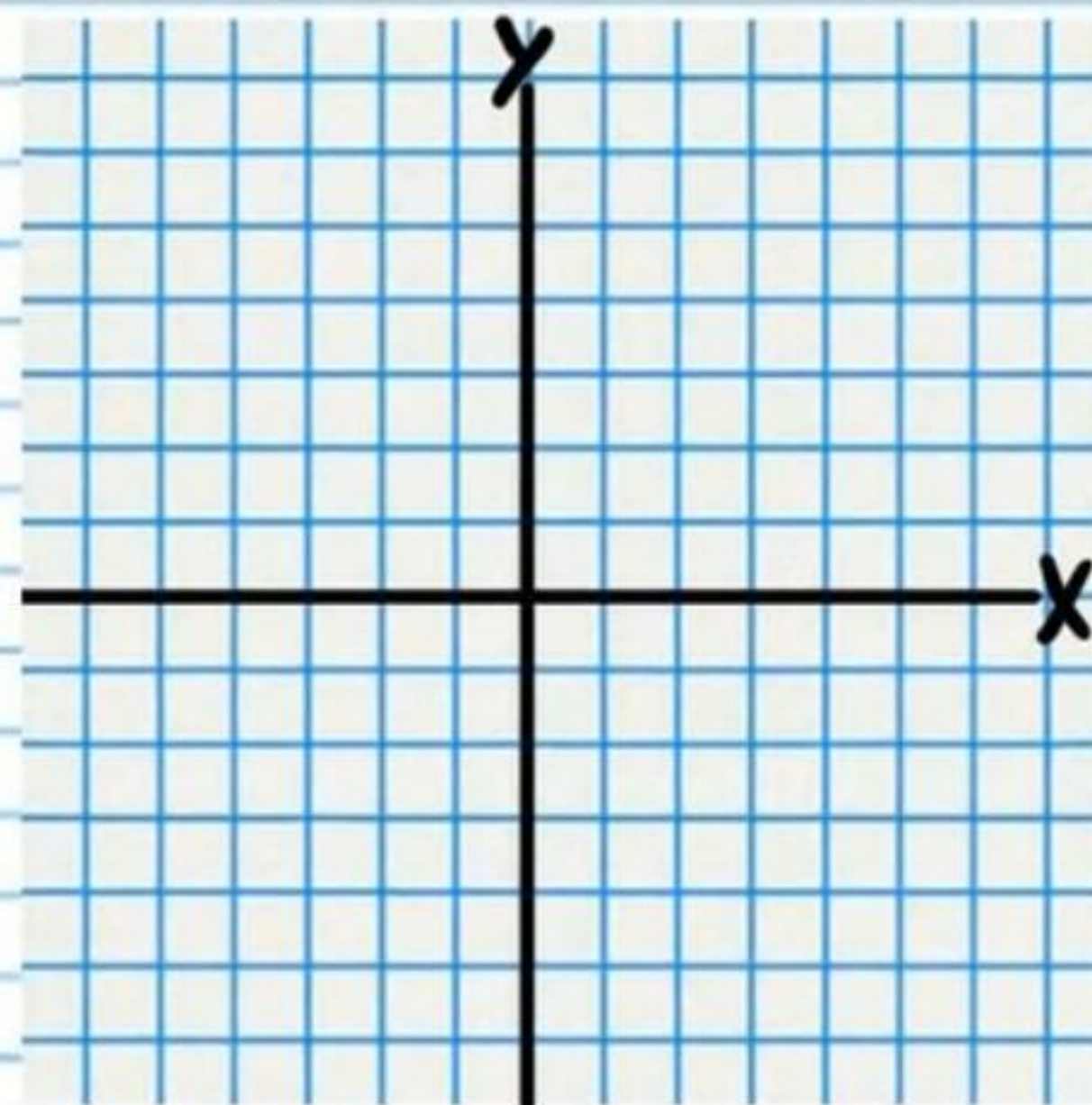


②③ Write the parent relation and the parent equation of the equation below. Then, graph the PARENT relation.

Parent Relation:

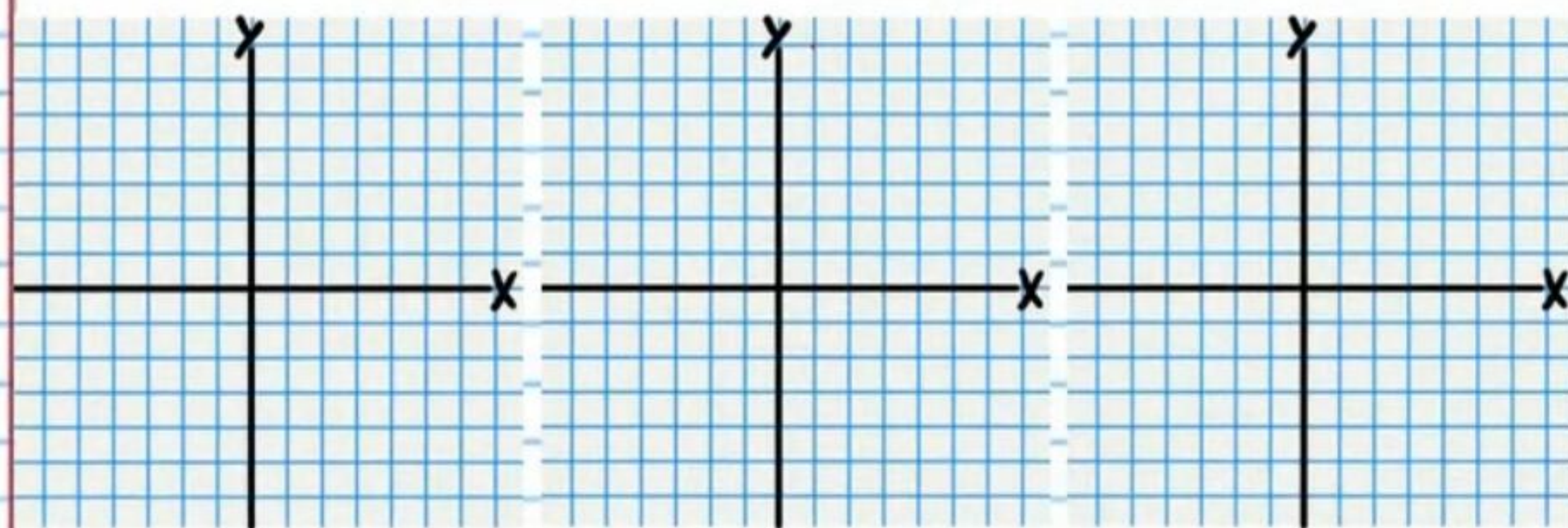
Parent Equation:

$$y = |-2x - 4| + 2$$



②④ Graph the equation above by performing multiple transformations on the parent graph.

Transformation #1 Transformation #2 Final Graph

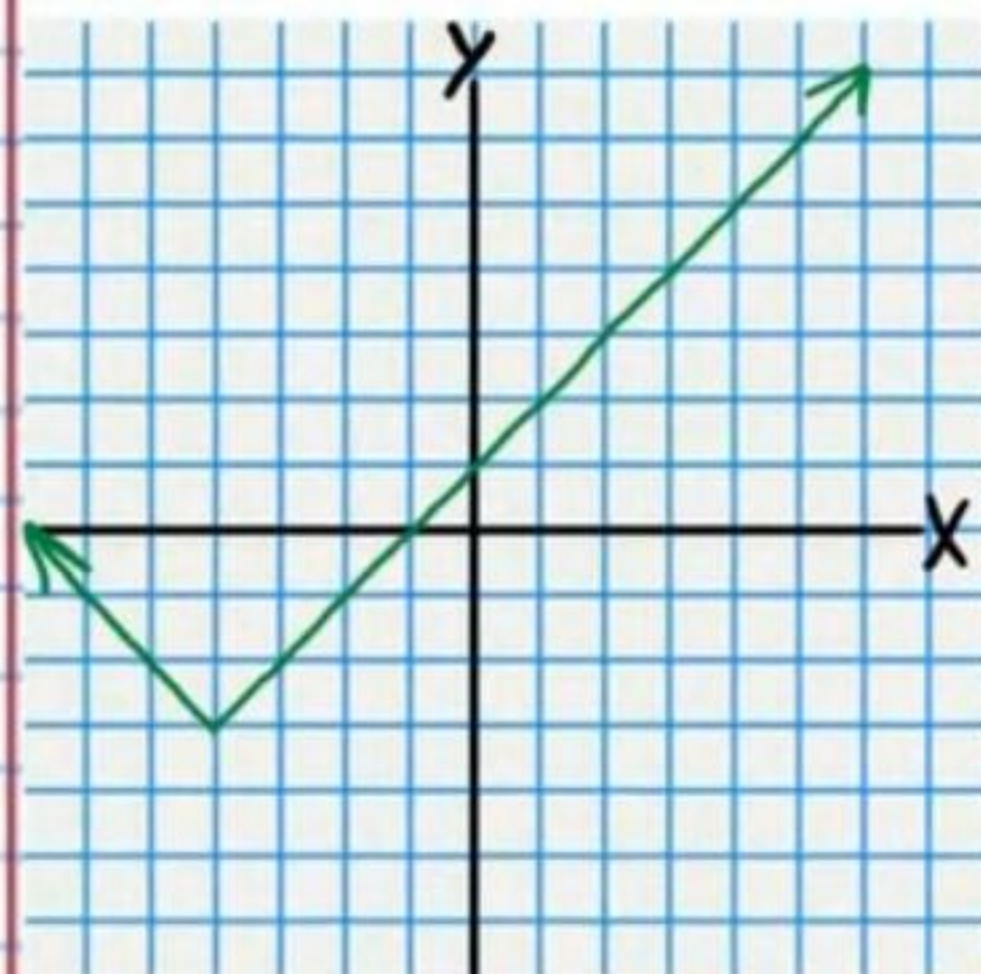


②⑤ Carrie found a bag containing quarters, nickels, and dimes. There are 59 coins and the total value of the coins is \$5.75. The number of quarters is 21 less than the number of nickels. How many of each type of coin are there?

Answers

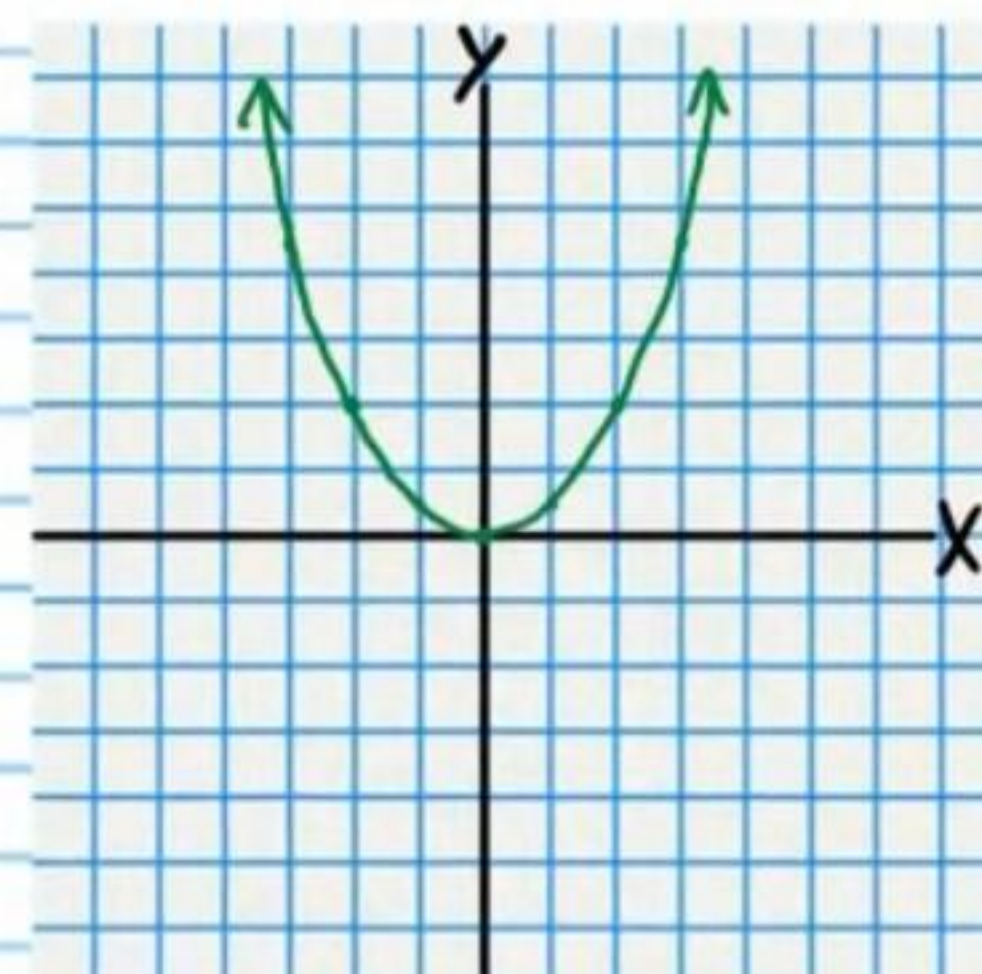
① Absolute Value

$$y = |x|$$



② Quadratic

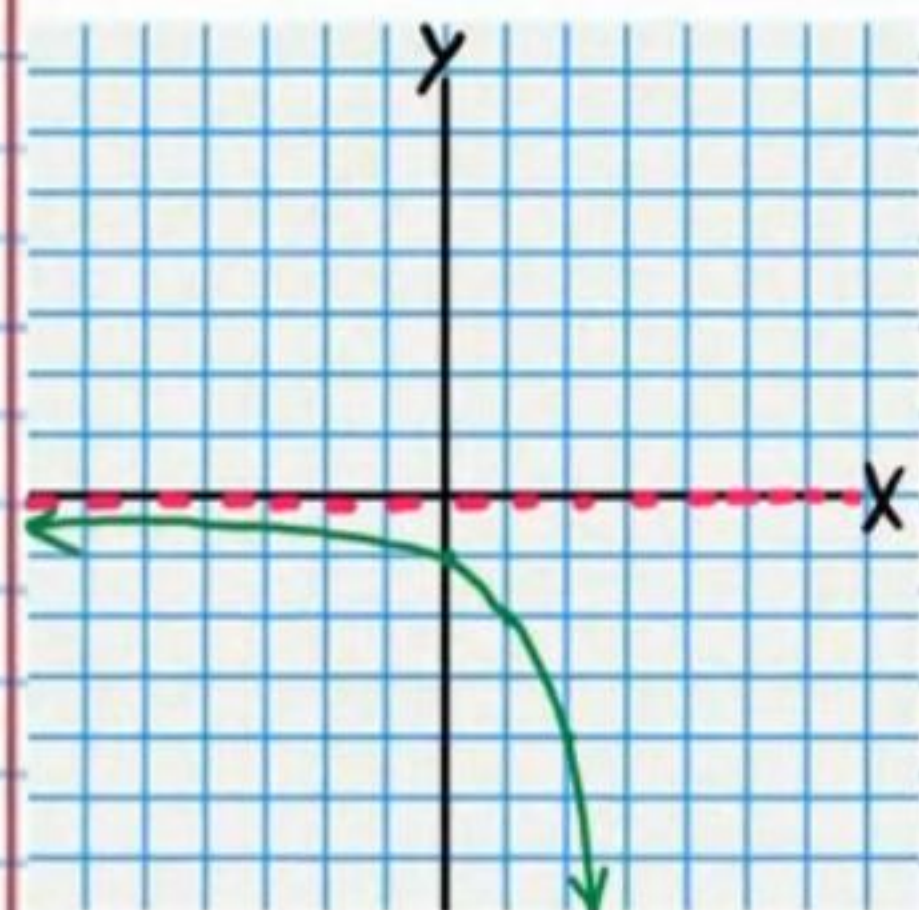
$$x^2 = y$$



③ Exponential

$$y = 2^x$$

$$y = 0$$

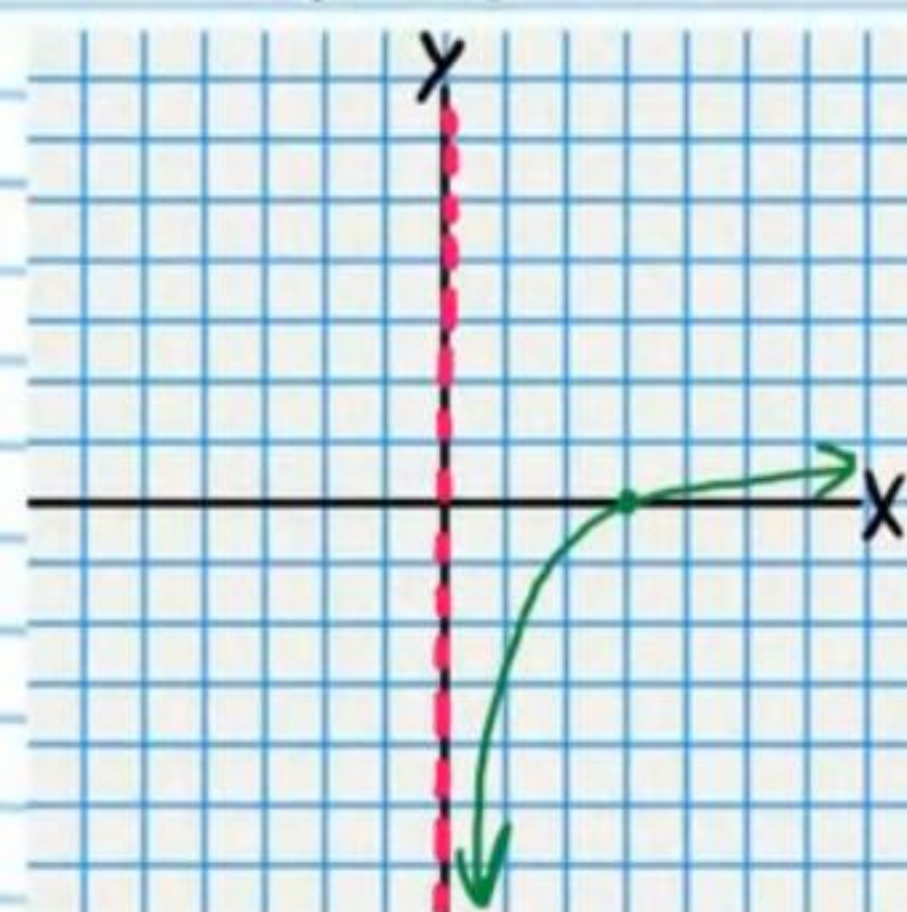


④ Logarithmic

$$\log_4 X = y$$

Horizontal stretch

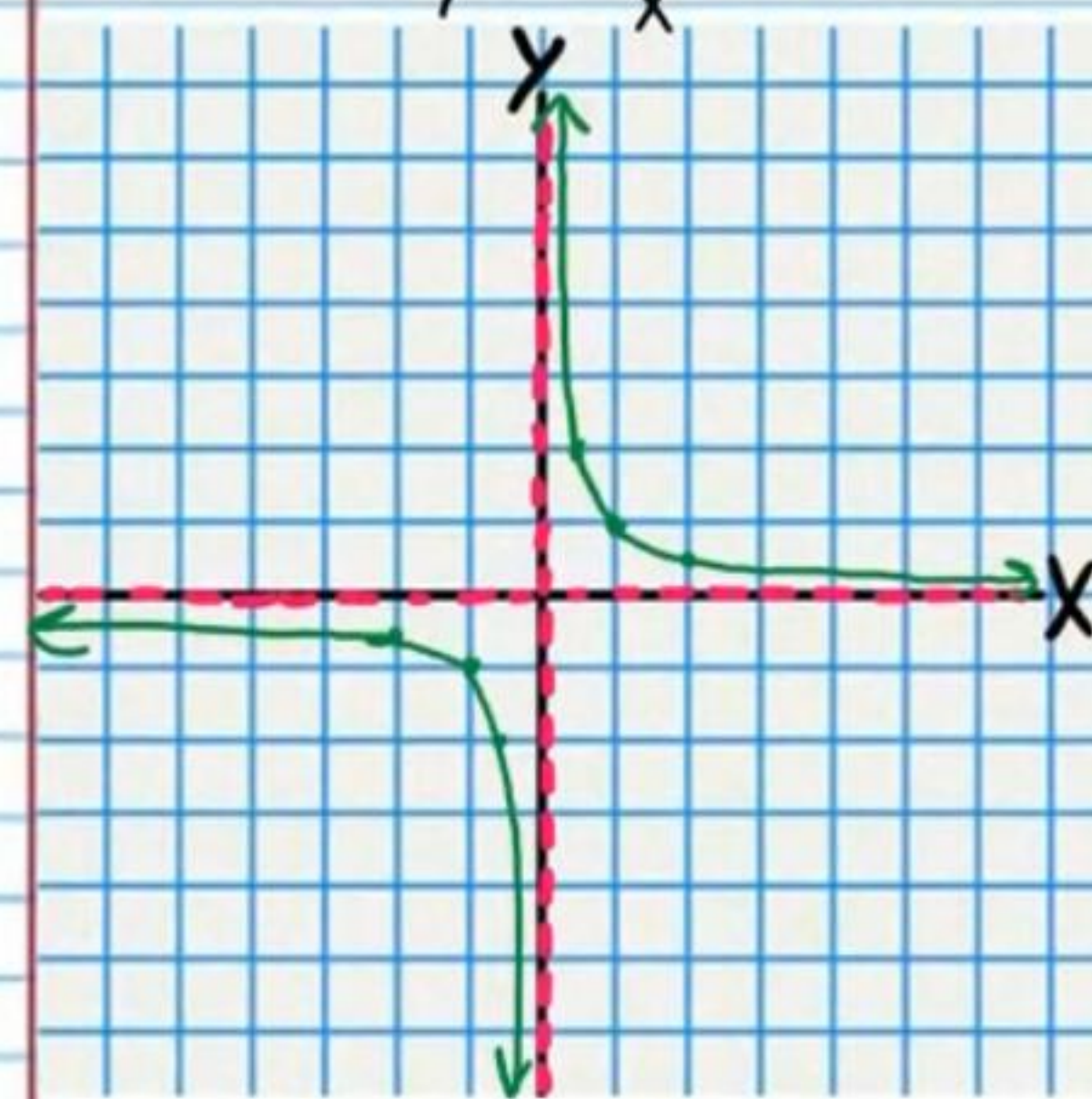
$$X = 0$$



⑤ $y = \frac{1}{-x}$

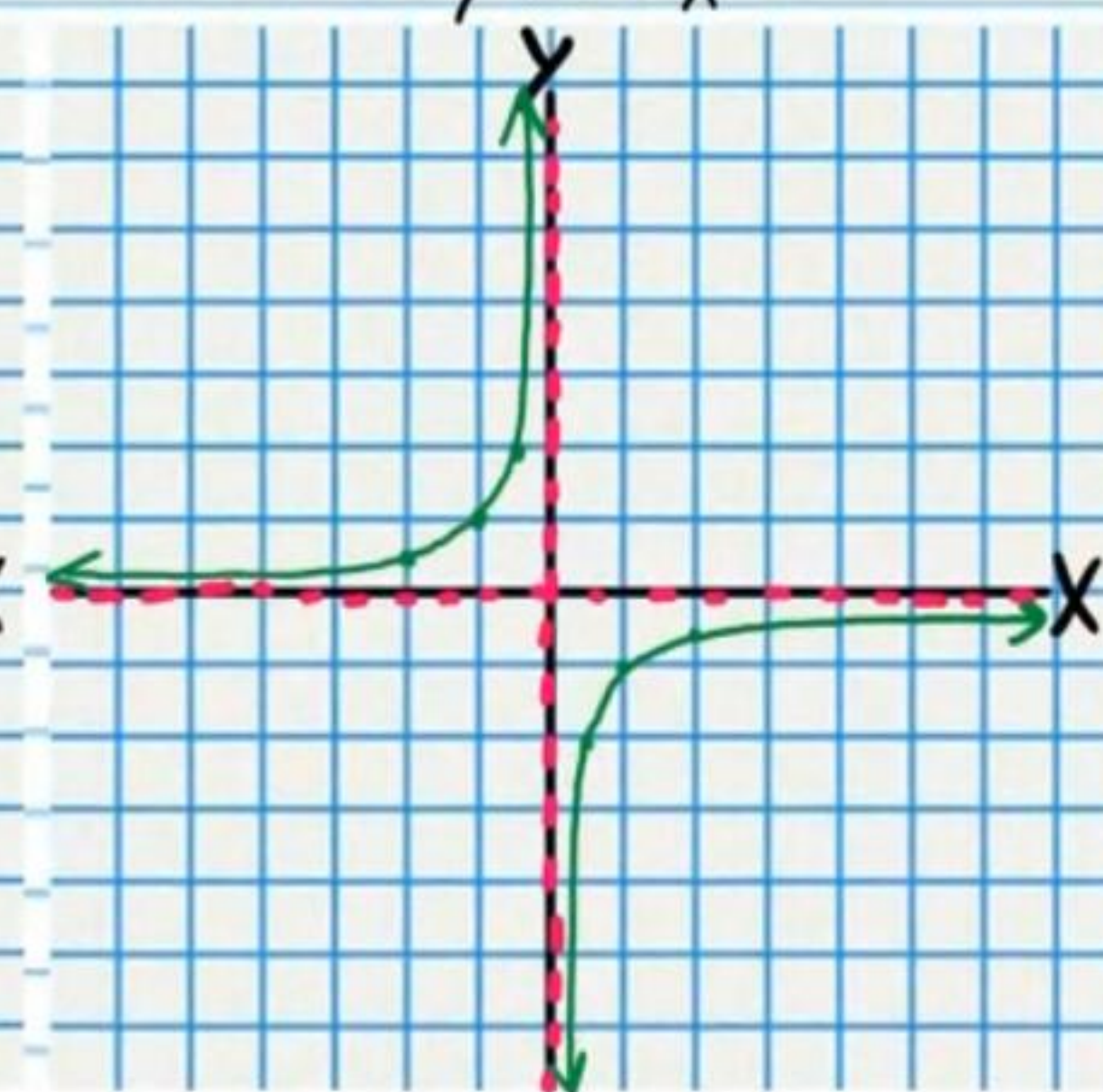
Parent Relation

$$y = \frac{1}{x}$$



Transformation

$$y = \frac{1}{-x}$$

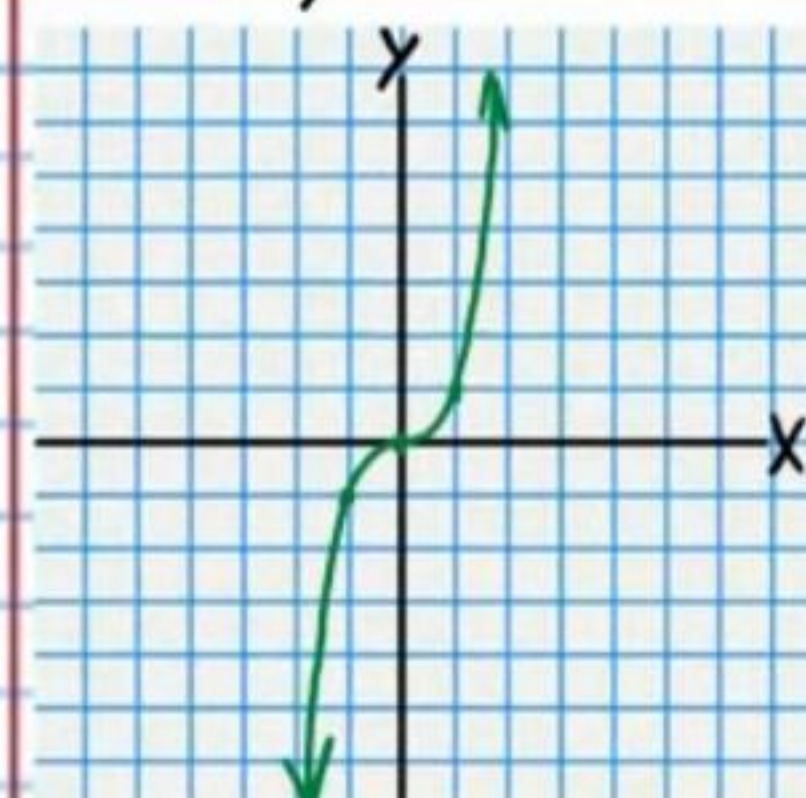


⑥ $-\frac{y}{4} = x^3$

Option I

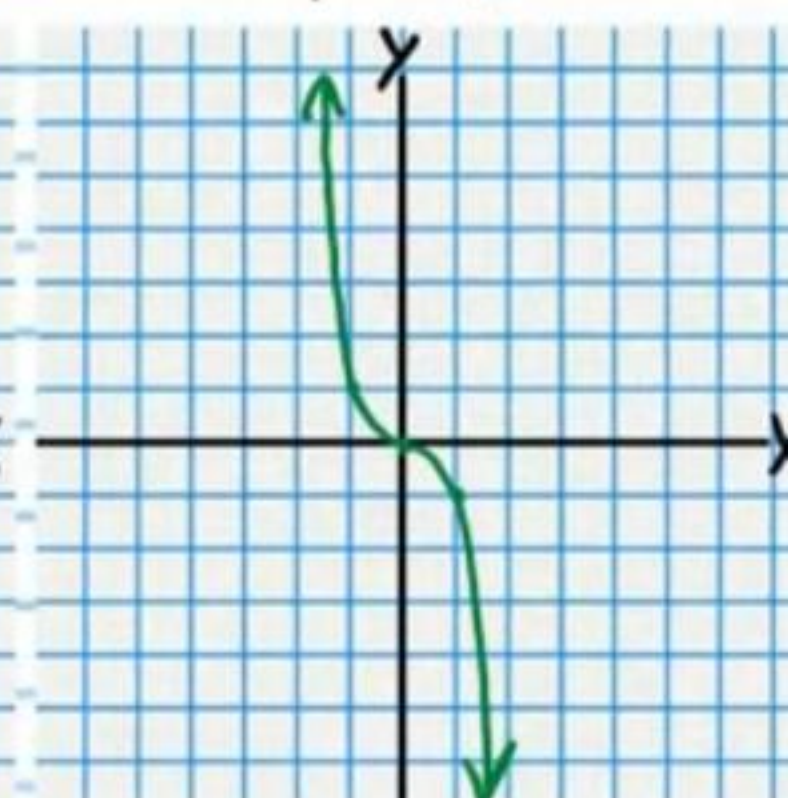
Parent Relation

$$y = x^3$$



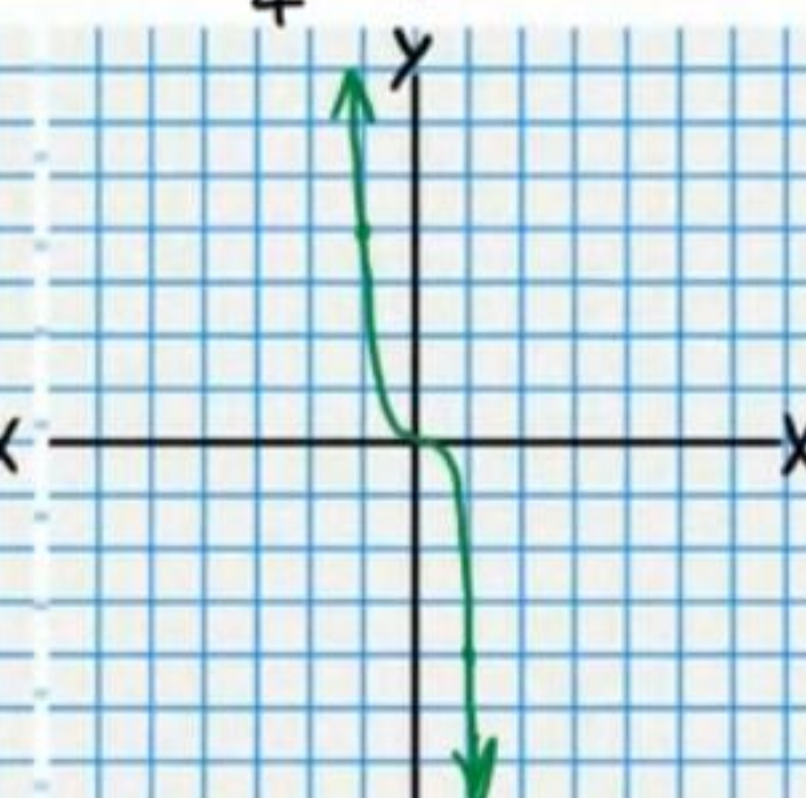
Transformation #1

$$-y = x^3$$



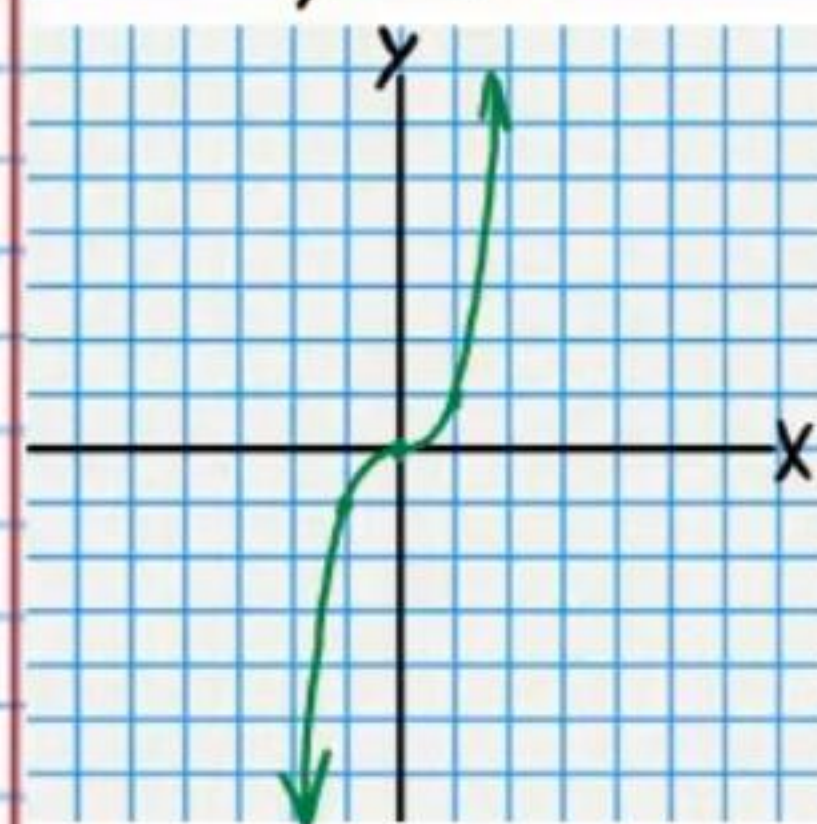
Transformation #2

$$-\frac{y}{4} = x^3$$



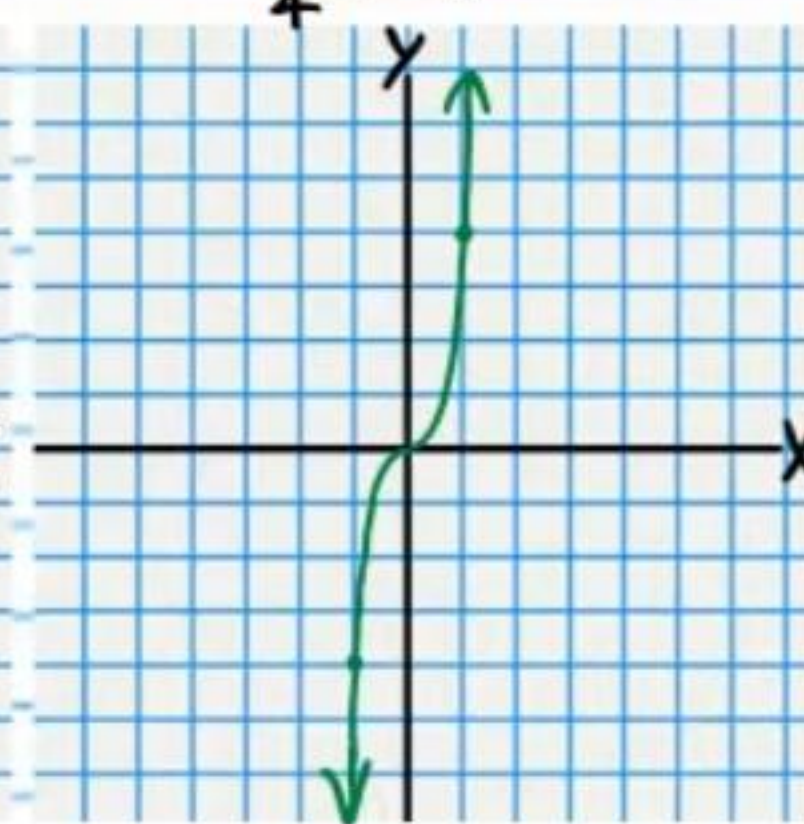
Option II

Parent Relation
 $y = x^3$



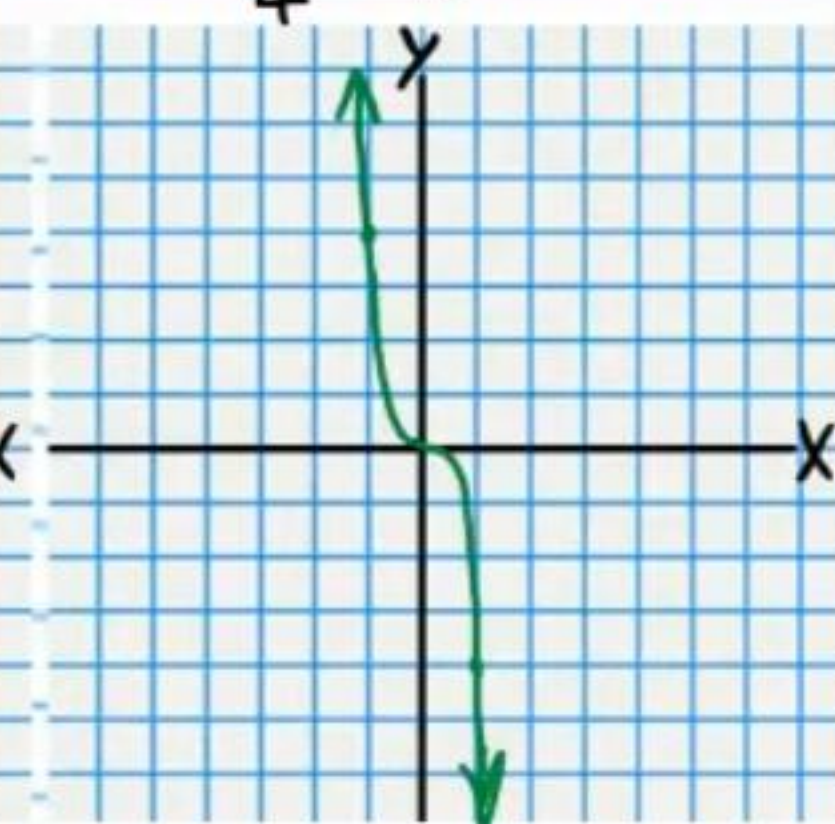
Transformation #1

$$\frac{y}{4} = x^3$$



Transformation #2

$$-\frac{y}{4} = x^3$$



⑦

x	y
-2	4
-1	5
0	10
1	12
2	9
2	-12

x	y
8	8
5	5
-2	14
0	5
2	2
8	7

x	y
3	15
2	10
1	8
0	10
-1	12
-2	14

x	y
7	0
10	0
9	0
4	0
2	4
1	87

⑧

x	f(x)
-1	0
1	$\sqrt{2}$
2	3
3	$2\sqrt{7}$

⑨ i) $\boxed{2}$ ii) $\boxed{0}$ iii) $\boxed{\text{Undefined}}$ iv) $\boxed{2\frac{1}{3}}$

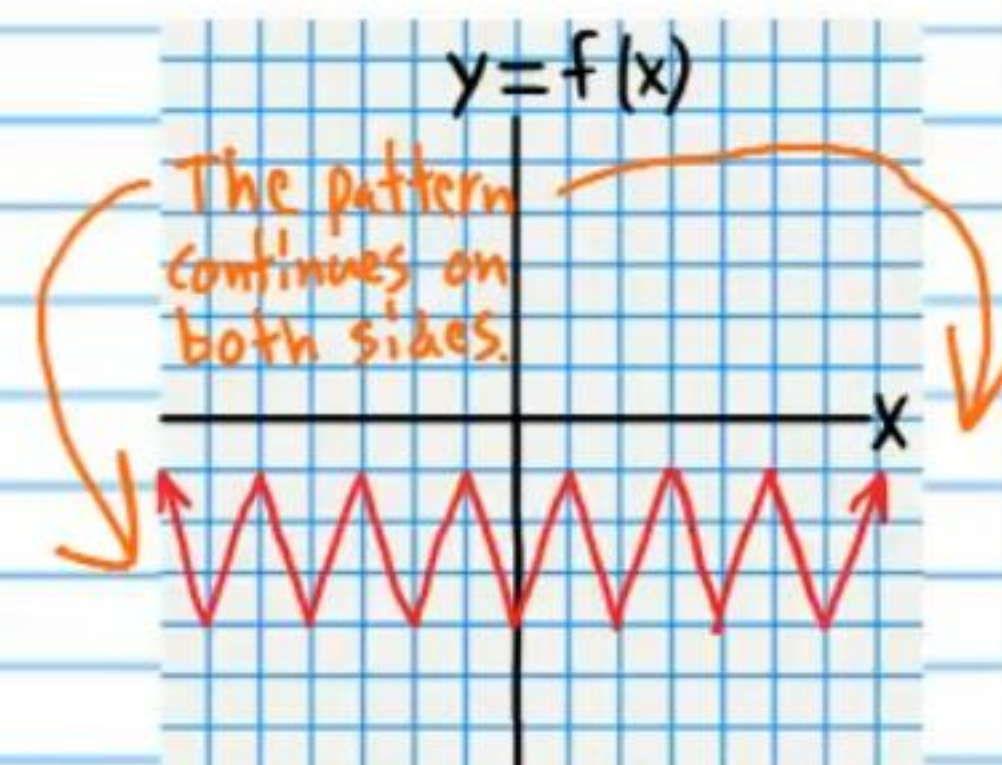
⑩ i) $\boxed{\log_2 X - \sqrt{X}}$ ii) $\boxed{1}$

⑪ $\boxed{(-5, 2) \cup [8, 100]}$

⑫ $\boxed{\{x \mid -5 < x < 2 \text{ or } 8 \leq x \leq 100\}}$

⑬ D: $\boxed{\text{All real numbers}}$

R: $\boxed{\{y \mid -4 \leq y \leq -13\}}$



⑭ $\boxed{2}$

⑮ $\boxed{\$672.87}$

⑯ $\boxed{14.02 \text{ years}}$

⑰ $\boxed{x \approx 0.183}$

⑱ $\boxed{x = 1,000,000 \text{ or } 10^6}$

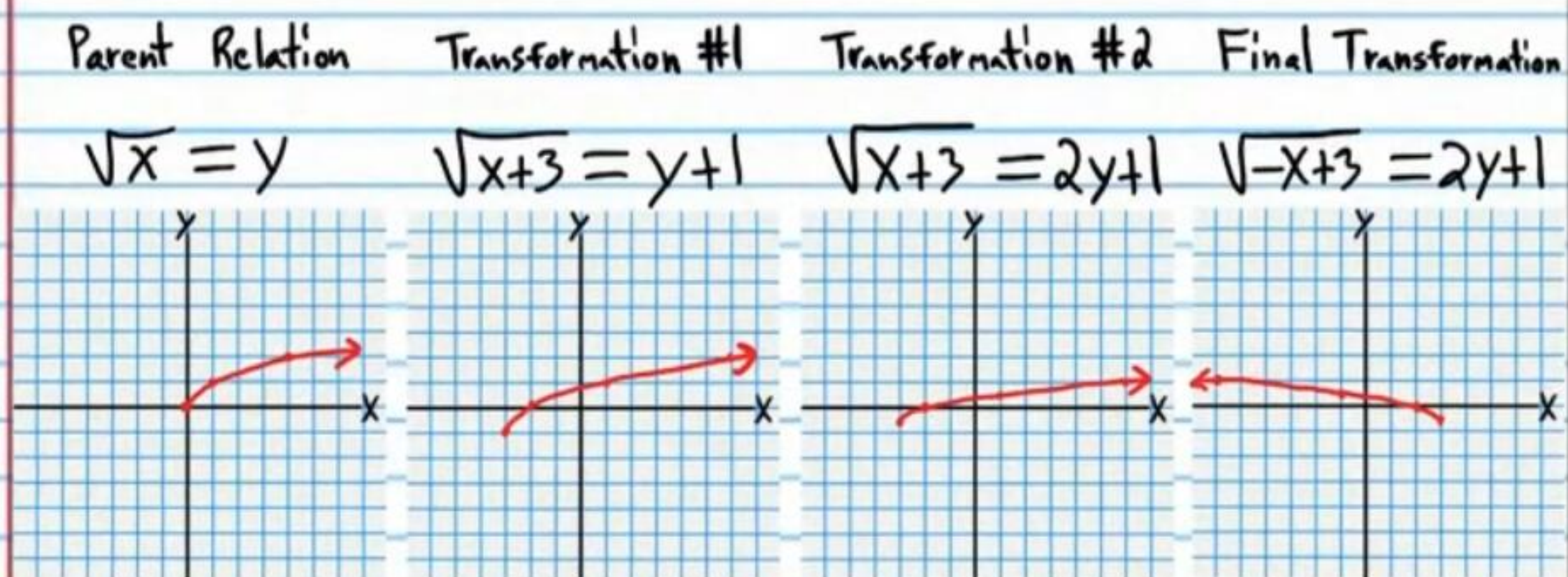
⑲ $\boxed{398.1 \text{ millimeters}}$

⑳ $\boxed{\{x \mid x > 13\}}$

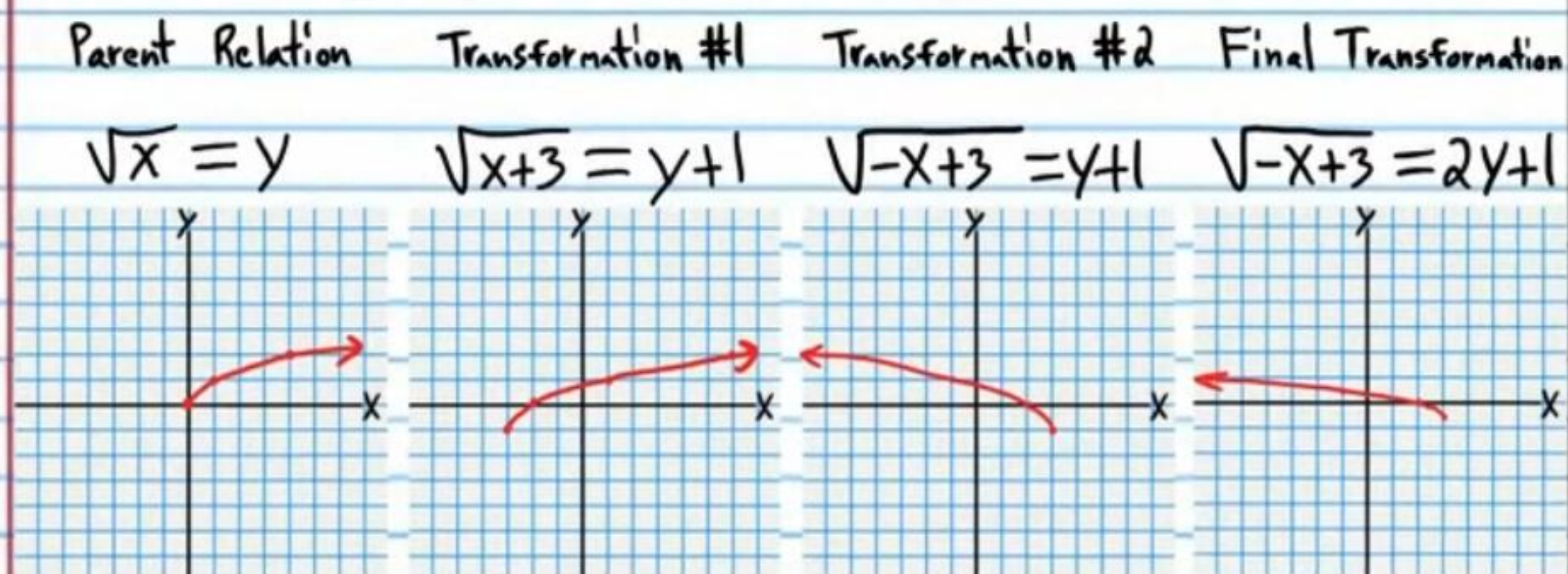
㉑ $\boxed{\text{All real numbers}}$

②② Pathway #1 or #2

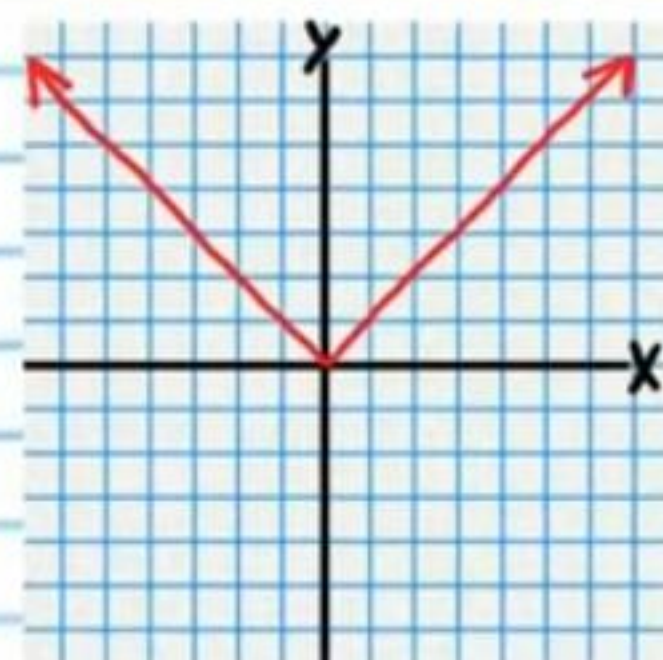
Pathway #1



Pathway #2

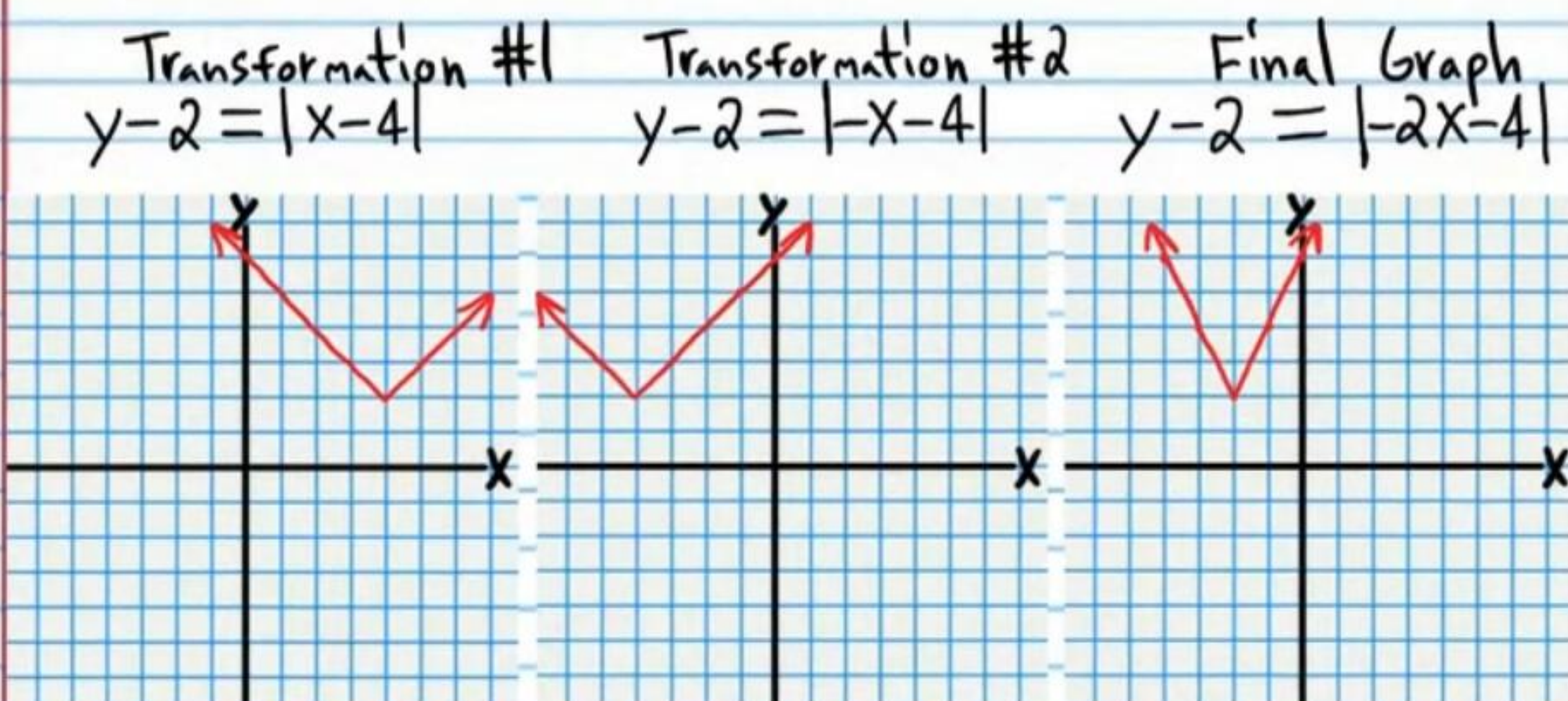


②③ Absolute Value
 $y = |x|$

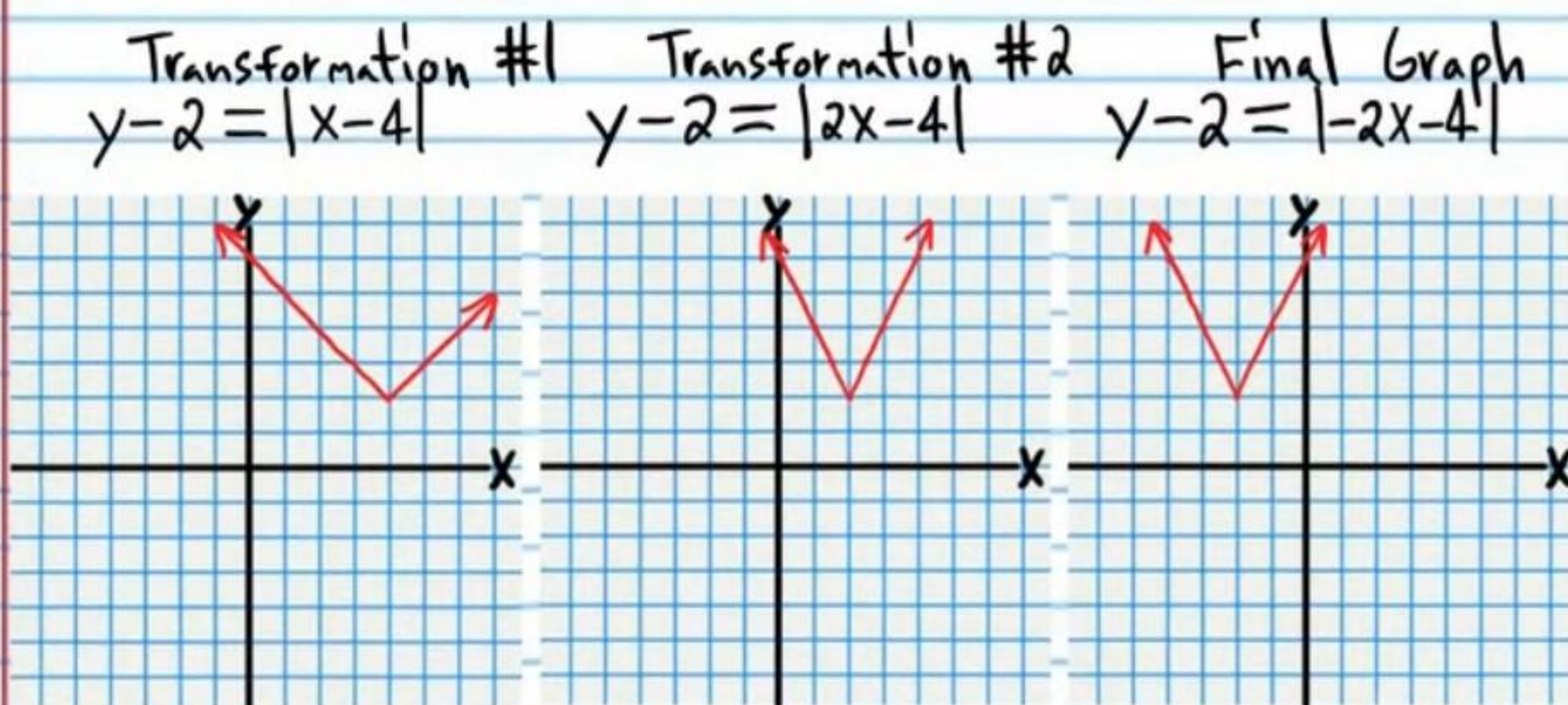


②④

Option #1



Optional #2



②⑤ $Q=9, D=20, N=30$